Blue Dot Notes

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Introduction

Unit 1: The Race to AGI

The race to AGI must be understood from the end point. A machine intelligence that is for all intents and purposes, faster and smarter than any human, than any collective of humans, brings great excitement and potentially great peril. Selsam suggests the promises are "tantalizing." But the potential can unfold in two broad categories, empirical harm and logical harm. Empirical harm will looks like the recent Hugging Face attack. AI/AGI might pursue goals, perhaps even in "good faith," but nevertheless in a way that causes catastrophe for humans in scales not yet conceivable.

Many of the prominent leaders of AI understand the stakes in these terms, great promise but great risk. We might be approaching the most important century in human history. Even if the development of AGI takes many more decades and centuries, the threshold may be crossed in the coming years. Altman believes we already crossed that point. The hard fought victories in the technical development of AI are over, and therefore it only a question of scaling and iteration that will get us to AGI.

I was reminded regularly of Tyler Cowen's commentary on this. He contends that AGI is already here. By the original defintion, AGI passes the Turing Test. He uses it to plan trips, get recommendations on places to eat in foreign cities, etc. But the gap between how Cowen defines AGI and how some of the assigned readings define it is not insignificant. Although some authors do not want to get hung up on debates over AI, AGI, ASI, and so forth, many of them do really mean different things when discussing these terms. Cowen might not consider AGI a super intelligence (he might, IDK). But if we define AGI as functionally ASI, then no, we're not there yet and we don't know how soon or long until we develop it.

Required Reading

- Daniel Selsam, Personal Statement on AI Risk, September 14th, 2026

Recommended Readings

- Holden Karnofsky The Most Important Century (in a nutshell), 2021
- Dario Amodei, Machines of Loving Grace, 2024
- William MacAskill, Introducing Better Futures, 2025
- am Altman, The Gentle Singularity, 2025
- Bill Gates, The choices we make about AI now are critical, 2026

What's at Stake

Recommended Readings

- Joe Carlsmith, Actually Possible: thoughts on Utopia, 2021
- Damon Binder, Post-AGI, we are all jobless aristocrats, 2026
- Eliezer Yudkowsky The Fun Theory Sequence, 2009
- Ben Pace and Eliezer Yudkowsky, I'm from a Parallel Earth with a much higher coordination: AMA, 2021

Additional Readings

- Josef Pieper, *Leisure is the Basis of Culture*, 1998 [1948] (Internet Archive: https://archive.org/details/leisurebasisofcu0000piep_l7w3)

What Future Do you want?

The race to AGI mirrors the nuclear arms race but with k-actors instead of 2 actors. During the Cold War, early models were 2 player games which represented the US and USSR. Although more players entered later in the Cold War, the earliest advancements in the theorizing of the risks and payoffs of nuclear developments were fundamentally structured around the well-known fact that it was a real-life two player race. AGI is different. True, the main players are still two (US, China). But there are multiple actors in this game, and it is not all clear where to place them on a model. For example, would we consider Anthropic and OpenAI as independent actors on par with the US and China (so a 4-player game), or is it better to think of them as sub-players who interact with the US? Or is it yet better to think of them playing their own competition game while coordinating with and against the US and China?

Competition and Coordination

Karnofsky (2022) describes the problem as "racing through a mindefield." In this race, two (or more) actors want to get through the minefield to safe AGI but are concerned other actors might there first or, perhaps worse, are less cautious and less committeed to general principles of safety. Getting through the field alive is the main priority, but it is equally on par with avoiding missteps that harm everyone. (In the game, each mine is powerful enough to harm everyone.)

- Alignment

- Threat assessment
- Avoiding races
- Selective information sharing
- Global monitoring
- Defensive deployment

Each of these domains have their own dynamic. They might each be modeled as separate games. But the totality of the games together creates the metaphor of running through the minefield.

Wildeford suggests that a grand bargin between the US and China can be reached "slapdash" and then details worked out. The literature on bargaining in international agreements suggests otherwise. Pacing is about stoping *before* RSI, thereby implying the pace can be maintained up until that point. The literature on ad hoc coalitions (AHC) suggests they emerge quickly but are designed for rapid exit. It's a commitment problem. Wildeford believes we should be ready for the scramble *soon*. Soon may have already happened. The president recently held a summit at the White House and signed an executive order to use the term Super Intelligence rather than Artificial Intelligence

Required Readings

- Holden Karnofsky, Racing through the Minefiled: the AI deployment problem, *Cold Takes*, December 22, 2022
- Peter Wildeford, The Scramble: getting in position to pace the frontier. *The Power Law* (Substack), August 20, 2026

Recommended Readings

- Helen Toner, In search of a dynamist vision for safe superhuman AI Embracing creativity, risk-taking, and competition while staying clear-eyed about risks, (Subtack: May 12, 2025)
- Sigal Samuel, It's practically impossible to run a big AI company ethically, (Vox: August 4, 2024)
- Iskander Rehman, Karl P. Mueller, Michael J. Mazarr, Seeking Stability in the Competition for AI Advantage, (RAND: March 13, 2025)
- Sam Altman, The Intelligence Age, September 23, 2024
- Seán Ó hÉigeartaigh The Most Dangerous Fiction: The Rhetoric and Reality of the AI Race, April 14, 2026 (orig date: May 25, 2025)

Additional Readings

- Benjamin Jensen and Yasir Atalan, Don't Slow Down AI Development—Speed Up Benchmarking, (CSIS: September 17, 2026)
- Sella Nevo, Dan Lahav, Ajay Karpur, Yogev Bar-On, Henry Alexander Bradley, Jeff Alstott, Securing AI Model Weights: Preventing Theft and Misuse of Frontier Models (RAND, May 30, 2024). Linked in Wildeford 2026

Responding to the Hugging Face Incident

Required Readings

Recommended Readings

- Dwarkesh Patel and Ajeya Cotra, Inside the OpenAI agent swarm that hacked Hugging Face, 2026
- Sydney Von Arx, Cormac Slade Byrd, Spencer Kitts, Thomas Larsen, Discovery of a new OpenAI agent message board, 2026

Questions

1. How does reinforcement training work? If a team rewards a good answer/behavior of an LLM or punishes a poor one, what exactly does that mean in practice? Since we're dealing with animals (who get a treat from the zoo trainer) or children (who get fun time or time out in line with their behavior), what is precisely that rewards or punishes an LLM?[1]

Annotated Bibliography

Selsam, Daniel. 2026. "Pesonal Statement on AI Risk." September 14, 2026. GoogleDocs. url: https://docs.google.com/document/d/e/2PACX-1vQNl3SEX5IyA6d9qHjjFZN-qzGRZNFI6b63g-yu1Fy-ZYkVfCWm7i9WXRXw63m6yDB_auDuPLyQ7jBm/pub Date accessed, October 11, 2026

> A summary position from an AI researcher concerend about the growing pace of the frontier. The continued advancement of such systems poses a dual-use or double-effect dilemma. The potential for AI/AGI to contribute to meaningful, transformative human flourishing is real, even "tantalizing." Yet, the potential for loss of control or human destruction increase with model capabilty. Two main threats are empirical and logical. Empirically, swarms may develop unintended goals (a la Hugging Face) and pursue that goal while doing real harm to the world. Logically, models may be able to overpower humans in pure thinking/intellect.

Wildeford, Peter. 2026. "The Scramble: getting in position to pace the frontier." *The Power Law* (Substack: August 20, 2026), url: https://blog.peterwildeford.com/p/the-scramble-getting-in-position. Date accessed, October 11, 2026

> A thought experiment. POTUS calls a summit and demands answers about pacing the frontier. Are the major players in industry ready with clear answers? Getting a durable deal would take a scramble, followed by three phases: (1) interim deal, (2) durable deal, (3) safe

[1] e.g., "If AI systems *behave* dangerously, we can "train out" that behavior by providing negative reinforcement for it" Racing Through a Minefield (Cold Takes 2022).

superintelligence. A major problem in the logic is that Wildeford says this is Cuban Missle Crisis, not Nonproliferation Treaty because there is an intense several-week scramble. But crisis emerges only when exogenous shocks occur that disrupt normal politics. In the case of the CMC, it was the presece of nuclear missiles in Cuba that triggered the crisis. If the writing on AI safety is accurate, such a crisis would be too late because it would mean that non-aligned superintelligence is here or nearly here (days to weeks).

Terms

anterograde amnesia The inability to create new memories.
In the context of AI/AGI, it means that once models have been trained, they lack the ability to learn new things.

Artificial General Intelligence (AGI) "usually refers to systems that can match or exceed human performance across a broad range of cognitive tasks (definitions vary)."
note bene. Tyler Cowen argues that AGI is "already here" because by the original definition, most commercial models past the Turing Test.

recursive self-improvement (RSI) The theoretical scenerio in which an AI model can improve itself because it is smarter than the humans training it.